Russia Ukraine-War's Impact on the Cost of a Healthy Diet

2026-08-10

Introduction

This study examines whether the Russia–Ukraine war increased the cost of a healthy diet in Egypt relative to a synthetic counterfactual constructed from comparable middle-income countries. Using the Synthetic Control Method (SCM), the analysis estimates how healthy diet costs in Egypt might have evolved in the absence of the shock.

Data Preparation

```
# rename dataset
world_development_data <- world_development_data %>%
  rename(
    country = `Country Name`,
    country_code = `Country Code`,
    year = `Time`,
    time_code = `Time Code`,
    gdp = `GDP per capita, PPP (constant 2021 international $)`,
    inflation = `Inflation, GDP deflator (annual %)`,
    exchange_rate = `Official exchange rate (LCU per US$, period average)`,
    agriculture_forestry_fishing_gdp = `Agriculture, forestry, and fishing, value added (% of GDP)`,
    trade_gdp = `Trade (% of GDP)`,
    population = `Population, total`,
    urban_population = `Urban population (% of total population)`,
    poverty_headcount = `Poverty headcount ratio at societal poverty line (% of population)`,
    unemployment_percentage = `Unemployment, total (% of total labor force) (national estimate)`
  ) %>%
  mutate(across(everything(), ~ifelse(. == "..", NA, .)))

# remove columns

# rename Egypt to match between two datasets
world_development_data <- world_development_data %>%
  mutate(country = recode(country,
                          "Egypt, Arab Rep." = "Egypt"))

# remove unnecessary columns and columns with too many missing values
world_development_data <- world_development_data %>%
  select(-country_code, -time_code, -poverty_headcount, -unemployment_percentage)

# remove remove unnecessary columns and columns with too many missing values from food prices data
```

```r
food_prices_data <- food_prices_data %>%
  select(-country_code, -cost_vegetables_ppp_usd, -cost_fruits_ppp_usd, -total_food_components_cost, -c
```

```r
# merge both dataset
data <- left_join(food_prices_data, world_development_data, by = c("country", "year"))
```

```r
# countries selected for control and treatment
selected_countries_data <- data %>%
  filter(country %in% c("India", "Indonesia", "Philippines", "Mexico", "Brazil", "Kazakhstan", "South A
    c(cost_healthy_diet_ppp_usd,
      annual_cost_healthy_diet_usd,
      gdp,
      inflation,
      exchange_rate,
      agriculture_forestry_fishing_gdp,
      trade_gdp,
      population,
      urban_population),
    as.numeric
  ))
```

```r
# Global variables

predictor_vars <- c(
      "gdp",
      "inflation",
      "exchange_rate",
      "agriculture_forestry_fishing_gdp",
      "trade_gdp",
      "population",
      "urban_population"
    )

outcome_var <- "cost_healthy_diet_ppp_usd"

special_predictors <- list(
    list(outcome_var, 2017, "mean"),
    list(outcome_var, 2018, "mean"),
    list(outcome_var, 2019, "mean"),
    list(outcome_var, 2020, "mean"),
    list(outcome_var, 2021, "mean")
    )

treatment_year <- 2022
pre_period <- 2017:2021
analysis_period <- 2017:2024
pre_idx <- which(analysis_period < treatment_year)
post_idx <- which(analysis_period >= treatment_year)
```

Descriptive Statistics

```
# descriptive statistics

pre_selected_countries_data <- selected_countries_data %>%
  filter(year < treatment_year)

desc_table <- pre_selected_countries_data %>%
  summarise(
    across(
      c(cost_healthy_diet_ppp_usd,
        gdp,
        inflation,
        exchange_rate,
        agriculture_forestry_fishing_gdp,
        trade_gdp,
        population,
        urban_population),
      list(
        Observations = ~sum(!is.na(.)),
        Mean = ~mean(., na.rm = TRUE),
        SD = ~sd(., na.rm = TRUE),
        Minimum = ~min(., na.rm = TRUE),
        Maximum = ~max(., na.rm = TRUE)
      )
    )
  ) %>%
  pivot_longer(
    everything(),
    names_to = c("Variable", "Statistic"),
    names_sep = "_(?=[^_]+$)"
  ) %>%
  pivot_wider(
    names_from = Statistic,
    values_from = value
  )

desc_table <- desc_table %>%
  mutate(across(where(is.numeric), ~ round(.x, 3)))

desc_table
```

```
## # A tibble: 8 x 6
##   Variable                          Observations     Mean       SD Minimum Maximum
##   <chr>                                    <dbl>    <dbl>    <dbl>   <dbl>   <dbl>
## 1 cost_healthy_diet_ppp_usd                   60   3.35e0 5.62e-1  2.12e0  4.75e0
## 2 gdp                                         60   1.76e4 7.33e+3  7.12e3  3.33e4
## 3 inflation                                   60   5.21e0 5.04e+0 -1.29e0  2.95e1
## 4 exchange_rate                               60   1.50e3 3.95e+3  3.19e0  1.46e4
## 5 agriculture_forestry_fishing_gdp            60   8.11e0 4.10e+0  1.95e0  1.87e1
## 6 trade_gdp                                   60   6.12e1 3.04e+1  2.43e1  1.34e2
## 7 population                                  60   2.06e8 3.67e+8  1.87e7  1.41e9
## 8 urban_population                            60   6.36e1 1.61e+1  3.32e1  8.71e1
```

```r
# Add country id

selected_countries_data$country_id <-
  as.numeric(factor(selected_countries_data$country))

selected_countries_data <- as.data.frame(selected_countries_data)

# Get Egypt's country ID
treatment_id <- unique(
  selected_countries_data$country_id[
    selected_countries_data$country == "Egypt"
  ]
)
```

Synthetic Control Analysis

```r
# Construct a synthetic version of Egypt using a weighted
# combination of donor countries that minimizes
# pre-treatment prediction error.

dataprep_out <-
  dataprep(
    foo = selected_countries_data,
    predictors = predictor_vars,
    dependent = outcome_var,
    special.predictors = special_predictors,
    unit.variable = "country_id",
    time.variable = "year",
    treatment.identifier = treatment_id,
    controls.identifier = setdiff(unique(selected_countries_data$country_id), c(treatment_id)),
    time.predictors.prior = pre_period,
    time.optimize.ssr = pre_period,
    unit.names.variable = "country",
    time.plot = analysis_period
  )

synth_out <- synth(dataprep_out)
```

```r
# Check predictor and donor country weights

synth_tables <- synth.tab(
  dataprep.res = dataprep_out,
  synth.res = synth_out
)

# Donor countries weight
synth_tables$tab.w
```

```
##    w.weights   unit.names unit.numbers
## 1      0.000       Brazil            1
```

```
## 2      0.000     Colombia            2
## 4      0.000        India            4
## 5      0.645    Indonesia            5
## 6      0.000   Kazakhstan            6
## 7      0.000     Malaysia            7
## 8      0.000       Mexico            8
## 9      0.000         Peru            9
## 10     0.003  Philippines           10
## 11     0.060 South Africa           11
## 12     0.292     Thailand           12
```

```
# Predictor weight
synth_tables$tab.v
```

```
##                                       v.weights
## gdp                                       0.088
## inflation                                 0.016
## exchange_rate                             0.005
## agriculture_forestry_fishing_gdp          0.089
## trade_gdp                                 0.049
## population                                0.137
## urban_population                          0.071
## special.cost_healthy_diet_ppp_usd.2017    0.098
## special.cost_healthy_diet_ppp_usd.2018    0.161
## special.cost_healthy_diet_ppp_usd.2019    0.145
## special.cost_healthy_diet_ppp_usd.2020    0.079
## special.cost_healthy_diet_ppp_usd.2021    0.062
```

```
# Compare predictors before and after constructing synthetic control

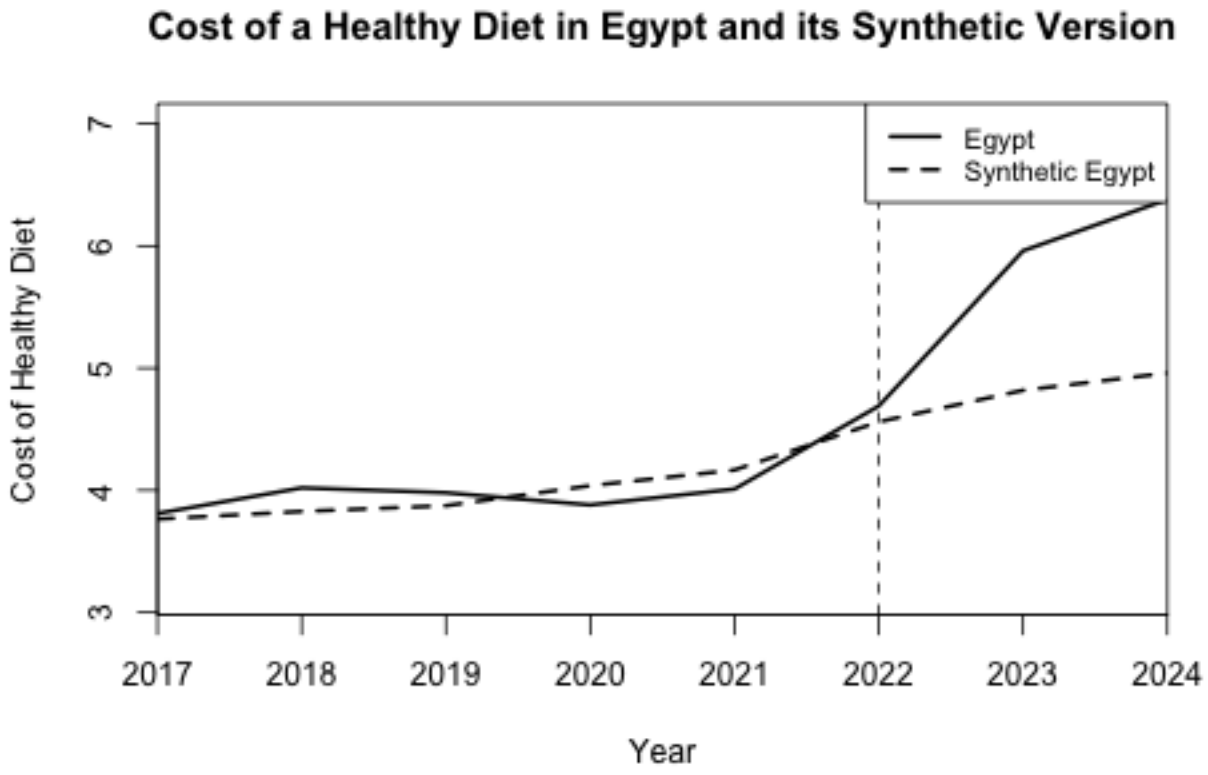
round(synth_tables$tab.pred,2)
```

```
##                                          Treated    Synthetic  Sample Mean
## gdp                                     14884.28     14847.60     17881.04
## inflation                                  15.07         2.58         4.32
## exchange_rate                              16.75      9126.94      1639.48
## agriculture_forestry_fishing_gdp           11.02        11.10         7.84
## trade_gdp                                  38.37        61.40        63.23
## population                             107440688.20 200411088.43 215427953.93
## urban_population                           42.49        55.97        65.56
## special.cost_healthy_diet_ppp_usd.2017      3.81         3.77         3.12
## special.cost_healthy_diet_ppp_usd.2018      4.02         3.83         3.16
## special.cost_healthy_diet_ppp_usd.2019      3.98         3.87         3.25
## special.cost_healthy_diet_ppp_usd.2020      3.88         4.04         3.40
## special.cost_healthy_diet_ppp_usd.2021      4.01         4.17         3.56
```

```
# Plot Cost of a Healthy Diet in Egypt and its synthetic version

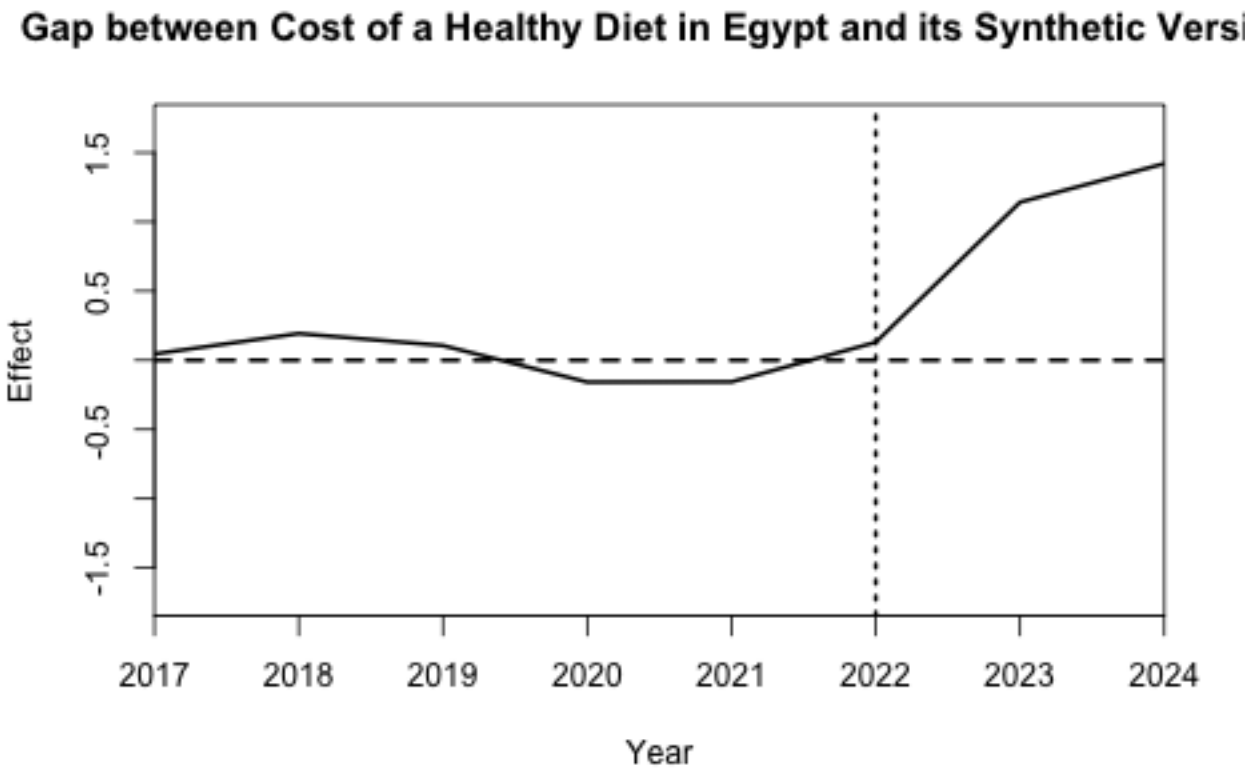
path.plot(
  dataprep.res = dataprep_out,
  synth.res = synth_out,
  Ylab = "Cost of Healthy Diet",
```

```
  Xlab = "Year",
  Legend = c("Egypt", "Synthetic Egypt"),
  abline(v=treatment_year, lty=2),
  Main = "Cost of a Healthy Diet in Egypt and its Synthetic Version"
)
```




```
# Plot gap between Cost of a Healthy Diet in Egypt and its synthetic version

gaps.plot(synth.res = synth_out,
          dataprep.res = dataprep_out,
          tr.intake = treatment_year,
          Ylab = "Effect",
          Xlab = "Year",
          Main = "Gap between Cost of a Healthy Diet in Egypt and its Synthetic Version")
```




```
# Egypt gap
egypt_gap <- dataprep_out$Y1plot -
  dataprep_out$Y0plot %*% synth_out$solution.w

egypt_gap
```

```
##                3
## 2017  0.04549571
## 2018  0.19311446
## 2019  0.10620286
## 2020 -0.15754695
## 2021 -0.15599944
## 2022  0.12990933
## 2023  1.14174787
## 2024  1.42057638
```

```
# Store gaps for placebo tests
store <- matrix(
  NA,
  nrow = length(analysis_period),
  ncol = length(unique(selected_countries_data$country_id))
)

colnames(store) <- unique(selected_countries_data$country)
rownames(store) <- analysis_period
```

```r
# Put Egypt gap in the first column
store[, "Egypt"] <- egypt_gap
```

Placebo Tests

```r
# Run placebo test for all donor countries

# All countries except Egypt
placebo_ids <- setdiff(
  unique(selected_countries_data$country_id),
  treatment_id
)

for (placebo_id in placebo_ids) {

  placebo_country <- unique(
    selected_countries_data$country[
      selected_countries_data$country_id == placebo_id
    ]
  )

  print(paste("Running placebo:", placebo_country))

  placebo_dataprep <- dataprep(
    foo = selected_countries_data,
    predictors = predictor_vars,
    dependent = outcome_var,
    special.predictors = special_predictors,
    unit.variable = "country_id",
    time.variable = "year",
    treatment.identifier = placebo_id,
    # Exclude BOTH the placebo country and actual treated country (Egypt)
    controls.identifier = setdiff(
      unique(selected_countries_data$country_id),
      c(placebo_id, treatment_id)
    ),

    time.predictors.prior = pre_period,
    time.optimize.ssr = pre_period,

    unit.names.variable = "country",
    time.plot = analysis_period
  )

  placebo_synth <- synth(placebo_dataprep)

  placebo_gap <- placebo_dataprep$Y1plot -
    placebo_dataprep$Y0plot %*% placebo_synth$solution.w

  store[, placebo_country] <- placebo_gap
```

```
}
```

```
## [1] "Running placebo: Brazil"
## [1] "Running placebo: Colombia"
## [1] "Running placebo: India"
## [1] "Running placebo: Indonesia"
## [1] "Running placebo: Kazakhstan"
## [1] "Running placebo: Malaysia"
## [1] "Running placebo: Mexico"
## [1] "Running placebo: Peru"
## [1] "Running placebo: Philippines"
## [1] "Running placebo: South Africa"
## [1] "Running placebo: Thailand"
```

```
# Plot placebo graph
placebo_gap <- store

# calculate MSE
mse <- apply(placebo_gap[pre_idx,]^2, 2, mean)

egypt.mse <- mse["Egypt"]

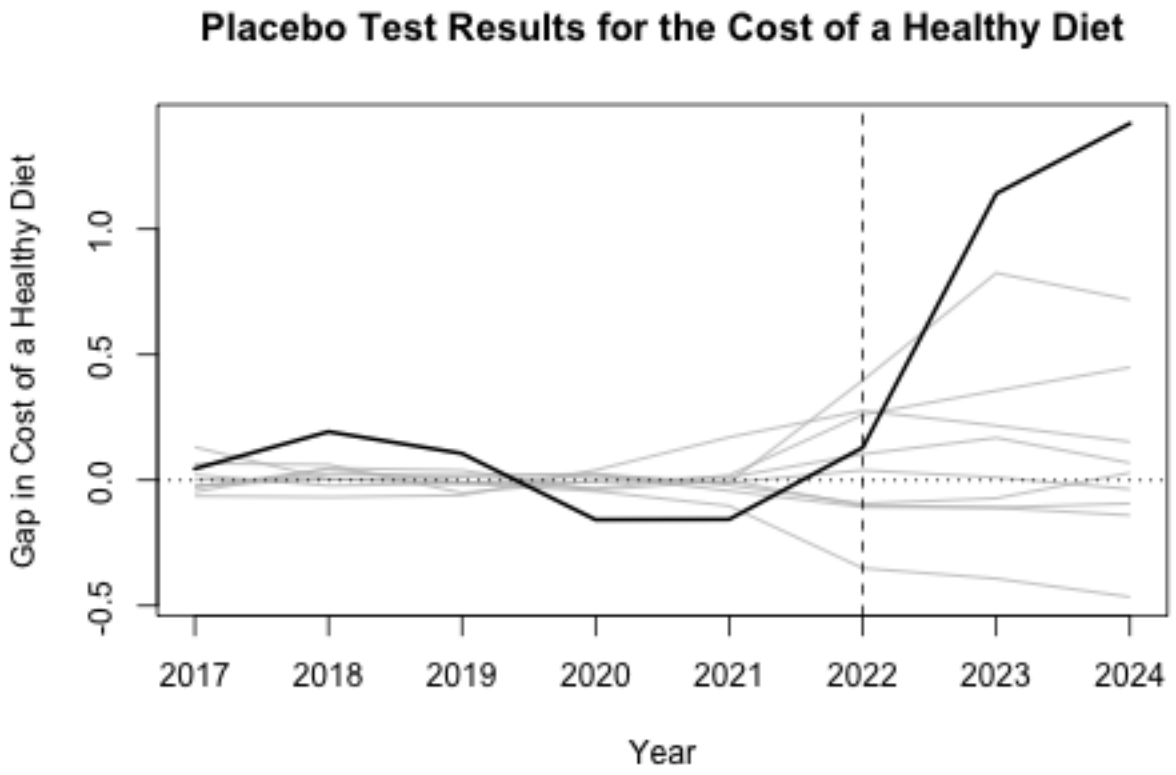
# remove placebo units with pre-treatment MSE more than five times that of Egypt
placebo_gap <- placebo_gap[, mse < 5 * egypt.mse]

plot(
  analysis_period,
  placebo_gap[, "Egypt"],
  type = "n",
  ylim = range(placebo_gap, na.rm = TRUE),
  xlab = "Year",
  ylab = "Gap in Cost of a Healthy Diet",
  main = "Placebo Test Results for the Cost of a Healthy Diet"
)

for(i in setdiff(
  1:ncol(placebo_gap),
  which(colnames(placebo_gap) == "Egypt")
)){
  lines(
    analysis_period,
    placebo_gap[, i],
    col = "gray"
  )
}

lines(
  analysis_period,
  placebo_gap[, "Egypt"],
  lwd = 2,
  col = "black"
)
```

```
abline(v = treatment_year, lty = 2)
abline(h = 0, lty = 3)
```




```
# Check pre and post RMSPE for all countries

mse <- apply(placebo_gap[pre_idx,]^2, 2, mean)

# Calculate pre-treatment RMSPE
pre_rmspe <- sqrt(
  apply(placebo_gap[pre_idx, ]^2, 2, mean)
)

# Calculate post-treatment RMSPE
post_rmspe <- sqrt(
  apply(placebo_gap[post_idx, ]^2, 2, mean)
)

# Calculate post/pre RMSPE ratio
rmspe_ratio <- post_rmspe / pre_rmspe

# Create summary table
rmspe_results <- data.frame(
  country = names(pre_rmspe),
  pre_rmspe = as.numeric(pre_rmspe),
  post_rmspe = as.numeric(post_rmspe),
```

```
  rmspe_ratio = as.numeric(rmspe_ratio)
)

rmspe_results
```

```
##         country  pre_rmspe post_rmspe rmspe_ratio
## 1        Brazil 0.09286515 0.22132278   2.3832706
## 2      Colombia 0.01018562 0.67207424  65.9826664
## 3         Egypt 0.14127913 1.05490946   7.4668454
## 4         India 0.01939687 0.07034701   3.6267208
## 5     Indonesia 0.04690632 0.03133828   0.6681036
## 6      Malaysia 0.02769291 0.11989595   4.3294812
## 7        Mexico 0.02836643 0.36372127  12.8222439
## 8          Peru 0.01600941 0.12092348   7.5532767
## 9   Philippines 0.03897642 0.10111589   2.5942836
## 10 South Africa 0.07734200 0.40552816   5.2433112
```

Sensitivity Analysis

```
# Run sensitivity: leave-one-out test

# Store results
loo_results <- data.frame()

# Loop through each donor to leave out
for (leave_out_id in placebo_ids) {

  # Name of country being left out
  leave_out_country <- unique(
    selected_countries_data$country[
      selected_countries_data$country_id == leave_out_id
    ]
  )

  cat("Leaving out:", leave_out_country, "\n")

  # Donor pool excluding treatment country AND the country being left out
  controls <- setdiff(
    unique(selected_countries_data$country_id),
    c(treatment_id, leave_out_id)
  )

  # Prepare data
  loo_dataprep <- dataprep(
    foo = selected_countries_data,
    predictors = predictor_vars,
    dependent = outcome_var,
    special.predictors = special_predictors,
    unit.variable = "country_id",
    time.variable = "year",
```

```
    treatment.identifier = treatment_id,
    controls.identifier = controls,
    time.predictors.prior = pre_period,
    time.optimize.ssr = pre_period,
    unit.names.variable = "country",
    time.plot = analysis_period
  )

  # Run synthetic control
  loo_synth <- synth(loo_dataprep)

  # Calculate gap: actual Egypt - synthetic Egypt
  gap <- loo_dataprep$Y1plot -
    loo_dataprep$Y0plot %*% loo_synth$solution.w

  # Pre-treatment RMSPE: 2017-2021
  pre_rmspe <- sqrt(
    mean(gap[pre_idx]^2)
  )

  # Post-treatment RMSPE: 2022-2024
  post_rmspe <- sqrt(
    mean(gap[post_idx]^2)
  )

  # Post/pre RMSPE ratio
  rmspe_ratio <- post_rmspe / pre_rmspe

  # Average post-treatment gap
  avg_post_gap <- mean(gap[post_idx])

  # Store results
  loo_results <- bind_rows(
    loo_results,
    data.frame(
      left_out_id = leave_out_id,
      left_out_country = leave_out_country,
      pre_rmspe = pre_rmspe,
      post_rmspe = post_rmspe,
      rmspe_ratio = rmspe_ratio,
      avg_post_gap = avg_post_gap
    )
  )
}
```

```
## Leaving out: Brazil
## Leaving out: Colombia
## Leaving out: India
## Leaving out: Indonesia
## Leaving out: Kazakhstan
## Leaving out: Malaysia
## Leaving out: Mexico
## Leaving out: Peru
```

```
## Leaving out: Philippines
## Leaving out: South Africa
## Leaving out: Thailand
```

```
loo_results
```

```
##    left_out_id left_out_country pre_rmspe post_rmspe rmspe_ratio avg_post_gap
## 1            1           Brazil 0.1408623   1.056558    7.500647    0.9000206
## 2            2         Colombia 0.1433206   1.055277    7.363053    0.8949199
## 3            4            India 0.1347123   1.051420    7.804927    0.9036670
## 4            5        Indonesia 0.1481730   1.049079    7.080094    0.9016332
## 5            6       Kazakhstan 0.1334036   1.048961    7.863058    0.9039903
## 6            7         Malaysia 0.1433209   1.055295    7.363164    0.8949427
## 7            8           Mexico 0.1420140   1.059389    7.459755    0.9015358
## 8            9             Peru 0.1412299   1.054755    7.468353    0.8972732
## 9           10      Philippines 0.1357638   1.068368    7.869313    0.9235244
## 10          11     South Africa 0.1426142   1.041120    7.300254    0.8823752
## 11          12         Thailand 0.2218561   1.226893    5.530132    1.0866670
```

Treatment Effects

```
# Gap and percentage effect between actual and synthetic Egypt by year

egypt_gap_table <- data.frame(
  year = analysis_period,
  actual_egypt = as.numeric(dataprep_out$Y1plot),
  synthetic_egypt = as.numeric(
    dataprep_out$Y0plot %*% synth_out$solution.w
  ),
  gap = as.numeric(egypt_gap)
) %>%
  mutate(
    percentage_effect = (gap / synthetic_egypt) * 100
  )

egypt_gap_table_display <- egypt_gap_table %>%
  mutate(
    actual_egypt = round(actual_egypt, 3),
    synthetic_egypt = round(synthetic_egypt, 3),
    gap = round(gap, 3),
    percentage_effect = round(percentage_effect, 2)
  )

egypt_gap_table_display
```

```
##   year actual_egypt synthetic_egypt    gap percentage_effect
## 1 2017         3.81           3.765  0.045              1.21
## 2 2018         4.02           3.827  0.193              5.05
## 3 2019         3.98           3.874  0.106              2.74
## 4 2020         3.88           4.038 -0.158             -3.90
```

```
## 5 2021         4.01           4.166 -0.156            -3.74
## 6 2022         4.69           4.560  0.130             2.85
## 7 2023         5.96           4.818  1.142            23.70
## 8 2024         6.38           4.959  1.421            28.64
```

```r
# Average post-treatment effects, 2022-2024

avg_post_gap <- mean(
  egypt_gap_table$gap[egypt_gap_table$year >= treatment_year]
)

avg_post_percentage_effect <- mean(
  egypt_gap_table$percentage_effect[
    egypt_gap_table$year >= treatment_year
  ]
)

cat(
  "Average post-treatment gap: $",
  round(avg_post_gap, 3),
  " per person per day\n"
)
```

```
## Average post-treatment gap: $ 0.897  per person per day
```

```r
cat(
  "Average post-treatment percentage effect:",
  round(avg_post_percentage_effect, 2),
  "%\n"
)
```

```
## Average post-treatment percentage effect: 18.4 %
```